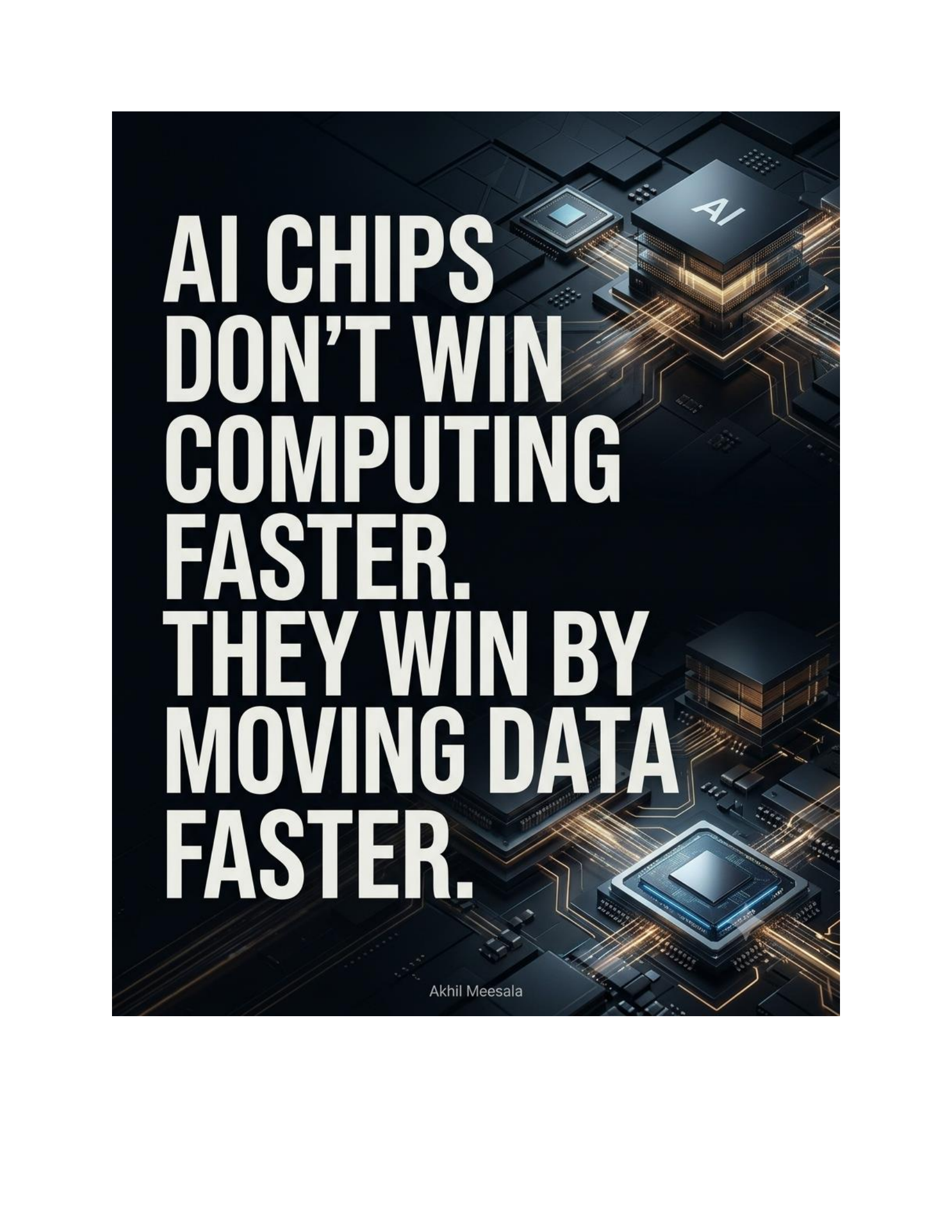


A detailed, high-tech illustration of a circuit board. Several AI chips are visible, with one prominently labeled 'AI' in white. The chips are interconnected by a network of glowing orange and yellow lines, representing data flow. The background is dark, emphasizing the intricate circuitry and the luminous connections.

**AI CHIPS
DON'T WIN
COMPUTING
FASTER.
THEY WIN BY
MOVING DATA
FASTER.**

Akhil Meesala

How AI Chips and GPUs Talk to Each Other: The Interconnect Architecture Behind Modern AI Systems

By Akhil Meesala | AI Chip Design & Semiconductor Architecture

When people think about AI computing, they usually focus on the processor.

More TOPS.
More FLOPS.
More memory.
More accelerators.

But there is another question that is becoming just as important:

How do all these AI processors communicate with each other?

A powerful AI accelerator sitting alone is useful.

A collection of GPUs, NPUs, CPUs, and custom AI ASICs working together is far more powerful—but only if they can exchange data fast enough.

That is where **high-speed interconnects, memory systems, chiplet architectures, and communication software** become critical.

The future of AI hardware isn't simply about building faster chips.

It is about building **faster systems of chips**.

The Basic Architecture

A modern AI system may contain:

CPU → GPU → AI Accelerator → Memory → Storage → Network

During AI training or inference, these components continuously exchange:

- Model weights
- Activations
- Gradients
- Tensors

- Intermediate results
- Metadata
- Control information

If the compute engines are extremely fast but the communication layer is slow, the accelerators spend valuable time waiting for data.

This creates a fundamental engineering problem:

Compute is only as fast as the system's ability to feed it data.

That is why interconnect architecture matters so much.

1. GPU-to-GPU Communication

The first layer is communication between GPUs.

In large AI systems, multiple GPUs may work together on the same model.

For example:

GPU 1 → GPU 2 → GPU 3 → GPU 4

They may need to exchange tensors during:

- Tensor parallelism
- Pipeline parallelism
- Data parallelism
- Collective operations
- Distributed inference

Traditional PCIe connectivity can become a bottleneck for extremely communication-intensive workloads.

This is why specialized high-bandwidth GPU interconnects exist.

NVIDIA's NVLink, for example, is designed specifically for high-bandwidth GPU-to-GPU communication. Current NVLink generations are designed to provide extremely high aggregate bandwidth and can connect GPUs through NVLink Switch fabrics for large-scale systems.

The principle is simple:

More compute → more data movement → more bandwidth required.

2. Chip-to-Chip Communication

Now consider a different scenario.

Instead of putting everything into one monolithic silicon die, designers can build a system from multiple dies or chiplets.

For example:

CPU Die + AI Accelerator Die + I/O Die + Memory Controller + Cache

These components need a high-speed communication fabric.

This is where technologies such as **chip-to-chip interconnects** become important.

NVIDIA NVLink-C2C is an example of a chip-to-chip interconnect designed for high-bandwidth, coherent communication between processors and custom silicon. It is also intended to support semi-custom systems and chiplet-based designs.

The important architectural idea is:

The processor doesn't have to be one giant piece of silicon.

The system can become modular.

3. Chiplets Change AI Chip Design

One of the most important trends in semiconductor architecture is the move from monolithic chips toward **chiplet-based systems**.

Instead of building:

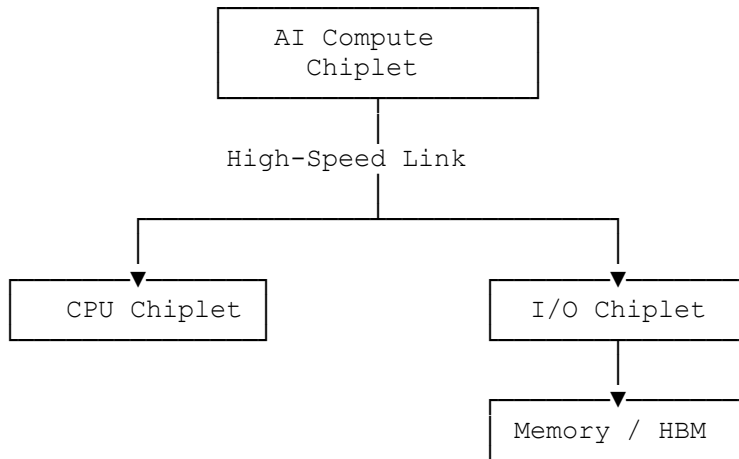
ONE HUGE DIE

we can build:

Chiplet A + Chiplet B + Chiplet C + Chiplet D

Each chiplet can specialize in a particular function.

For example:



This architecture can improve design flexibility, manufacturing economics, scalability, and reuse.

UCIe (Universal Chiplet Interconnect Express) is an open industry standard for die-to-die connectivity inside packages. Its specifications define the physical layer, protocols, software model, and interoperability mechanisms for chiplet-based systems.

The latest UCIe 3.0 specification supports data rates up to **64 GT/s**, reflecting the industry's push toward higher-bandwidth chiplet communication.

4. Does This Mean All Chips Share One Memory?

Not necessarily.

This is one of the most misunderstood concepts in AI hardware.

There is a difference between:

Data sharing

and

Memory coherency / shared memory semantics.

Two accelerators may exchange data through an interconnect without literally accessing one physical memory pool.

A system could have:

GPU 1 → HBM
GPU 2 → HBM
GPU 3 → HBM
GPU 4 → HBM

with high-speed communication between them.

Another architecture may provide a more coherent memory model.

For example, NVIDIA's NVLink-C2C can provide a coherent connection between CPUs and GPUs/custom silicon in supported architectures.

The architecture therefore depends on what the system is trying to optimize:

- Bandwidth
- Latency
- Memory capacity
- Coherency
- Power
- Cost
- Scalability

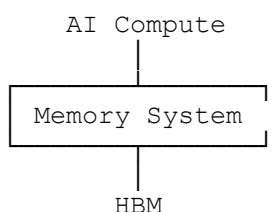
5. HBM Is Part of the Equation

Interconnects are only one side of the problem.

The AI accelerator also needs extremely high memory bandwidth.

This is why **High Bandwidth Memory (HBM)** has become so important for AI accelerators.

A simplified architecture looks like:



The objective is to keep the compute units continuously supplied with data.

This creates a hierarchy:

Registers → Cache → HBM → System Memory → Storage

Each layer has different:

- Capacity
- Latency
- Bandwidth
- Cost
- Power characteristics

Good AI-chip architecture is therefore largely about **moving the right data to the right place at the right time.**

6. The Interconnect Is More Than Wires

A high-speed interconnect is not simply a collection of physical wires.

It involves multiple layers.

Physical Layer

Handles the electrical signaling.

This includes:

- SerDes
- PHY
- Lanes
- Clocking
- Signal integrity
- Equalization

Link Layer

Responsible for reliable communication.

It can handle things such as:

- Flow control

- Error detection
- Retransmission
- Link management

Protocol Layer

Defines how information is interpreted.

Examples include:

- PCIe
- CXL
- NVLink
- UCIe-related protocols

UCIe itself defines a complete die-to-die interconnect architecture including physical-layer and protocol components, and can leverage PCIe and CXL standards.

Software Layer

And finally, software has to understand the hardware.

Drivers, runtimes, communication libraries, compilers, and distributed frameworks coordinate data movement and computation.

This is where hardware and software architecture meet.

7. Software Communication Matters

Imagine you have eight GPUs.

The hardware can provide extremely high bandwidth.

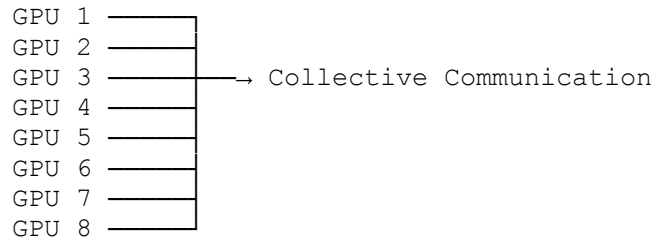
But if the software sends data inefficiently, the theoretical bandwidth doesn't matter.

AI workloads rely heavily on collective communication operations such as:

- All-reduce
- All-gather
- Reduce-scatter
- Broadcast
- Point-to-point communication

For distributed training, the communication pattern can become just as important as raw compute performance.

For example:



The goal is to minimize:

Communication time + synchronization overhead

while maximizing:

Useful compute time

8. Why Latency Matters

Bandwidth tells us:

How much data can I move?

Latency tells us:

How long does it take before communication begins to matter?

Both matter.

Consider two accelerators exchanging a small tensor repeatedly.

Even if the interconnect has enormous bandwidth, high latency can still hurt performance.

This is particularly important for:

- Small tensor operations
- Synchronization
- Distributed inference
- Control-heavy workloads
- Fine-grained communication

For large tensors, bandwidth becomes increasingly important.

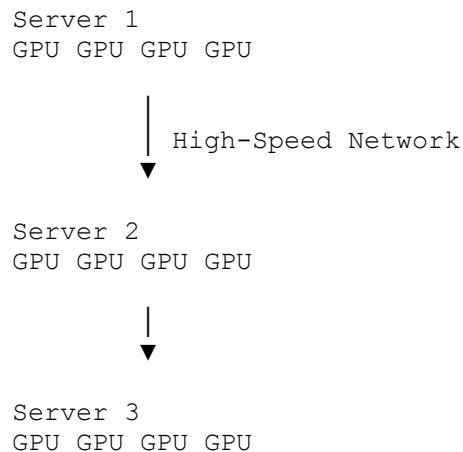
So AI-chip architects constantly balance:

Latency vs Bandwidth

9. Scaling Beyond One Server

Eventually, the architecture must move beyond a single server.

You may have:



Now we have another architectural layer:

Scale-up

Communication inside a server or accelerator domain.

Scale-out

Communication between servers.

The technologies and design trade-offs are different.

At large AI scale, both become critical.

Modern AI infrastructure therefore looks less like:

"A computer with a GPU"

and more like:

"A distributed computer made from thousands of communicating accelerators."

10. Why Custom AI ASICs Need Great Interconnects

This is especially important for companies designing custom AI accelerators.

Suppose you create an AI ASIC that delivers enormous compute performance.

But the accelerator cannot receive data quickly enough.

Your theoretical compute capability becomes underutilized.

This is the classic:

Compute-bound vs communication-bound problem.

A good AI ASIC therefore needs to be designed together with:

- Compute architecture
- Memory hierarchy
- HBM interface
- Cache architecture
- Interconnect
- DMA engines
- Network interface
- Software runtime
- Compiler
- Communication libraries

You cannot optimize the accelerator in isolation.

11. The Future: AI Systems Made of Specialized Silicon

The next generation of AI infrastructure is likely to become increasingly heterogeneous.

Instead of:

CPU + GPU

we may see:

CPU + GPU + NPU + AI ASIC + DPU + memory accelerators + networking accelerators

Each component performs a specialized job.

The interconnect becomes the nervous system connecting them.

This is why technologies such as NVLink-C2C and UCle are strategically important: they represent different approaches to connecting increasingly heterogeneous compute elements at high bandwidth and low latency.

12. The Bigger Picture

When designing an AI chip, the question shouldn't simply be:

"How many TOPS can this chip deliver?"

A better set of questions is:

How fast can we feed the compute units?

How efficiently can accelerators communicate?

How much memory bandwidth is available?

How much data must move between chips?

How much energy does communication consume?

Can the architecture scale from one accelerator to thousands?

These questions often determine whether an AI accelerator performs well in the real world.

Because AI performance is not just:

Compute

It is:

Conclusion

The future of AI hardware will not be defined by a single faster processor.

It will be defined by **systems of processors working together efficiently**.

High-speed interconnects provide the communication fabric.

HBM provides massive memory bandwidth.

Chiplet architectures provide modularity.

Coherent protocols provide more sophisticated memory and communication semantics.

Software runtimes coordinate the entire system.

And AI workloads determine how all of these pieces should work together.

The most powerful AI infrastructure may therefore look less like a traditional computer and more like a **networked system of specialized silicon**.

That is the direction AI chip architecture is moving toward.

And for anyone working in AI hardware, semiconductor design, chiplets, or accelerator architecture, one principle is becoming increasingly important:

Don't design the chip first and the system later. Design the compute, memory, interconnect, packaging, and software as one architecture.

— Akhil Meesala

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